

Intro Section

Motivation

Image processing is implemented into almost any sort of social media application, and there are lots of variations of software available, from free and open-source alternatives, like GIMP, to enterprise software that is locked behind a monthly subscription, such as Adobe's suite offers. Each variation offers its own problems — the open source option is quite limiting and very hard to use — the learning curve associated is too great, and many users are discouraged by it. Adobe's suite, while rich in features and ease of use, is still very heavy, and, most importantly for my project's point of view, allows only for destructive alteration of the image.

Overview

This project is a lightweight, albeit simplified within the scope of the senior project, non-destructive image editor, which is easy to use and allows for version control — going back and forth from different alterations made to the image, and allowing for the merging of different versions. This allows for the free editing of the desired image without having to worry about having to revert all changes made — you can always branch out, remove undesired alterations and merge between different 'branches' of edits. Along with this, as is expected of an image editor, the application allows for the alteration of an imported image — color adjustment, gaussian blur brush, and dynamic resizing through seam carving. Color adjustment allows the precise control over exposure, the brightness of the image, highlights, i.e. controlling the brightness of the brightest tones and areas in the image, and shadows, the darkest areas of the image, where 'information' might be lost in the dark.

